

Comparison of Gas and Liquid Phase Methanol Synthesis Processes

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ABSTRACT

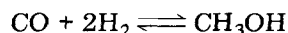
The synthesis of methanol from CO and H₂ has conventionally been carried out in a fixed-bed reactor in the gas phase. Recently, liquid phase methanol synthesis in a slurry reactor has received considerable attention, owing to the advantages of easy temperature control and the absence of diffusional resistances. In this work, the reactors used for gas phase and liquid phase methanol synthesis are analyzed. The fixed-bed reactor is modeled using a one-dimensional heterogeneous model while the slurry-bed reactor is modeled using a plug flow model for the gas phase and the axial dispersion model for the slurry phase. The physical and thermodynamic properties of the systems are estimated using literature correlations. The main assumptions underlying the model equations are stated and solution techniques described. For each type of reactor, the effects of various parameters such as temperature, pressure, phase velocities, diameter and height of reactor are investigated with the help of the simulators developed. Finally, a comparison of the performances of the fixed-bed and slurry-phase reactors is presented.

1. INTRODUCTION

Methanol is a basic industrial chemical that is produced at an annual rate of over ten million tons. Plant capacity for methanol

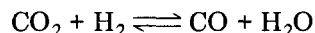
is increasing and there is the possibility that its use, chiefly as a fuel, will eventually require large additional amounts of methanol [1, 2].

Traditionally, methanol has been produced by catalytic hydrogenation of carbon monoxide



$$\Delta H_{298}^\circ = -94\,084 \text{ kJ kmol}^{-1} \quad (\text{reaction 1})$$

An important side reaction, the hydrogenation of CO₂ (reverse shift reaction) also takes place under synthesis conditions



$$\Delta H_{298}^\circ = 41\,270 \text{ kJ kmol}^{-1} \quad (\text{reaction 2})$$

For the main reaction (reaction 1), the conversion achievable is greatly limited by thermodynamic equilibrium and since the reaction is exothermic and involves a contraction in volume, the highest yields and conversions of methanol are obtained at high pressure and low temperatures. Several generations of catalysts have been developed to get a reasonable reaction rate. Zinc chromium oxide was one of the earliest used. Later a catalyst based on a mixed zinc, copper oxide supported on chromium oxide or aluminum oxide was introduced [3].

In 1966, ICI introduced their low pressure methanol process in low tonnage plants, taking advantage of a much more active catalyst. A low-pressure methanol synthesis process is advantageously combined with production of syngas by partial oxidation since the latter can be carried out at methanol synthesis pressure, thus avoiding the necessity of intermediate gas compression. Typical

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operating conditions for the process are: pressure, 5 - 10 MPa and temperature 220 - 280 °C. The reactor operates adiabatically and the temperature rise at each bed is known to be extremely high. For this reason, multitbed quench reactors are necessary. The overall effectiveness factor is known to be less than 0.5.

The liquid phase methanol process (LPMeOH) was evaluated against the conventional multitbed quench system by Sherwin and Frank [4]. Chem Systems Inc. has been developing the LPMeOH technology under the sponsorship of the Electric Power Research Institute (EPRI) since 1975 [5]. In 1981, Air Products and Chemicals, Inc. started a 42-month research and development program to prove the technical feasibility of LPMeOH [6]. In these two projects bubble column slurry reactors were used and several aspects of new technology were investigated. In the process, syngas containing CO, CO₂, and H₂ is passed upward into the bubble column slurry reactor concurrent with the inert hydrocarbon (Witco-40 or Freezene-100) which serves to both fluidize the catalyst and to absorb the exothermic heat of reaction. Owing to the easy temperature control and absence of diffusional resistances, this type of reactor has received considerable attention.

In this paper, both gas phase and liquid phase methanol syntheses are analyzed. Mathematical models for the simulation of fixed- and slurry-bed methanol synthesis reactors have been developed to make a comparison between these two types of operation. Model equations are derived and the solution techniques are included. For each type of reactor, the effects of various parameters on the performance are investigated with the help of the simulators developed. Finally a comparison of the reactors is presented. The physical and thermodynamic properties of the systems (gas and liquid phase reactor system) are estimated by the literature correlations and these methods are included in Appendix A.

2. MODEL DESCRIPTION

Owing to the distinct differences in the design and operation of fixed and slurry

reactor systems, the two analyses are presented separately. The main assumptions underlying the model equations are stated and solution techniques for the equations are described.

2.1. Fixed-bed reactor design for methanol synthesis

The fixed-bed reactor is the traditional type used to carry out methanol synthesis. It refers to two-phase systems in which the reacting gas flows through a bed of catalyst particles or pellets [7].

The design of fixed-bed reactors requires that both transport processes (mass and energy transfer) and the intrinsic kinetics are considered in the reactor design. The model equations should include transport restrictions against mass and heat transfer in the interior of the pellet and the gas film surrounding the particle.

The mathematical model which allows for differences in concentrations and temperature between the bulk gas stream and the particles is called the heterogeneous reactor model [8]. In the heterogeneous reactor model, the effects of the internal and external resistances on the reaction rate are accounted for by using global or overall rather than intrinsic rates. These types of rates are expressed in terms of gas phase conditions and they can be directly used in the axial integration of mass and energy balance differential equations.

2.1.1. Model assumptions

In the model development the following assumptions are made.

(i) The significance of axial dispersion depends on reactor length, effective diffusivity and gas velocity. For high velocities ($Re > 1$) and for reactors whose length to diameter ratio is well above unity, axial dispersion is negligible [9]. Therefore a plug flow model is reasonable.

(ii) Since radial gradients in concentration and temperature are caused solely by radial variation in axial velocity, and since in the plug flow case, the axial velocity is only dependent on the axial direction, radial profiles can be neglected in an adiabatic plug flow reactor [9].

(iii) Unlike high pressure synthesis, low pressure synthesis operates at relatively

moderate pressures and temperatures where the nonideality of the gas mixture can be neglected [10]. The rate expressions for low pressure synthesis to be used here are also based on the concentrations rather than fugacities, suggesting that the ideal gas law may be applicable.

(iv) The presence of two simultaneous reactions with very nonlinear rate equations makes the use of the Stefan–Maxwell equation for the mixture gas diffusivities difficult. Hence Wilke's equation will be used to calculate the diffusivities in the gas mixture [7].

In this work, we wish to present a model that is as general as possible. Therefore, the following points which were neglected in other studies (see, for example, Capelli *et al.* [11]) will be considered here.

(a) Both reactions, methanol formation and shift reaction, proceed with finite reaction rates and neither of them is considered to be at equilibrium.

(b) The temperature gradients inside the pellets can be very important. Hence non-isothermal effectiveness factors are considered in the solution of solid phase balances.

(c) Since the total number of moles change due to reaction 1, the concentration profiles should be affected. This effect is accounted for in gas phase material balances, but since it leads to a more complicated problem in the calculation of effective reaction rates it is assumed to be negligible in the solid phase.

(d) Physical properties are considered to be a function of temperature and pressure. Both heat and mass transfer parameters and also mixture properties are considered to vary along the length of the reactor.

2.1.2. Model equations

Based on the assumptions above, the following model equations can be written. In these equations methanol and CO_2 are chosen as key components.

Mass balances in the gas phase are represented as follows. For methanol

$$\frac{d}{dx}(UC_M^g) = (k_g)_M a_v \{(C_M^s)_{r=R} - C_M^g\} \quad (1)$$

and for carbon dioxide

$$\frac{d}{dx}(UC_{\text{CO}_2}^g) = (k_g)_{\text{CO}_2} a_v \{(C_{\text{CO}_2}^s)_{r=R} - C_{\text{CO}_2}^g\} \quad (2)$$

In the solid phase we write for methanol

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 D_M^e \frac{dC_M^s}{dr} \right) = -r_1 \rho_p \quad (3)$$

and for carbon dioxide

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 D_{\text{CO}_2}^e \frac{dC_{\text{CO}_2}^s}{dr} \right) = r_2 \rho_p \quad (4)$$

Here effective diffusivities are assumed constant at their bulk values.

Energy balances are represented in the gas phase by

$$\frac{d}{dx}(U\rho C_p T^g) = h a_v \{(T^s)_{r=R} - T^g\} \quad (5)$$

and in the solid phase by

$$\frac{\lambda_e}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = \{(-\Delta H_R)_1 r_1 + (-\Delta H_R)_2 r_2\} \rho_p \quad (6)$$

Boundary conditions are set as follows. For $z = 0$

$$C_i^g = C_{i,0}^g, \quad T^g = T_0^g, \quad U = U_0 \quad (7)$$

for $r = 0$

$$\frac{dC_M^s}{dr} = \frac{dC_{\text{CO}_2}^s}{dr} = \frac{dT^s}{dr} = 0 \quad (8)$$

while for $r = R$

$$(k_g)_M \{(C_M^s)_{r=R} - C_M^g\} = -D_M^e \left(\frac{dC_M^s}{dr} \right)_{r=R} \quad (9)$$

$$\begin{aligned} & (k_g)_{\text{CO}_2} \{(C_{\text{CO}_2}^s)_{r=R} - C_{\text{CO}_2}^g\} \\ & = -D_{\text{CO}_2}^e \left(\frac{dC_{\text{CO}_2}^s}{dr} \right)_{r=R} \end{aligned} \quad (10)$$

$$h \{(T^s)_{r=R} - T^g\} = -\lambda_e \left(\frac{dT^s}{dr} \right)_{r=R} \quad (11)$$

2.1.3. Solution technique

It is clear that integration of eqns. (1), (2) and (5) can easily be performed provided that the catalyst surface concentrations $(C^s)_{r=R}$ and temperature $(T^s)_{r=R}$ are known. These values are obtained by the solution of the solid phase mass and energy equations (3), (4) and (6) with the boundary conditions of eqns. (8) - (11). Numerical solution of the solid phase mass balance equations was carried out by the orthogonal collocation method. A computer program COLSYS [12] which uses B-spline collocation functions was

applied and solid phase concentration and temperature profiles were obtained. The gas phase concentration profiles, on the other hand, were evaluated by the solution of gas phase mass balance equations by the Runge-Kutta-Gill method [13].

2.2. Slurry-bed reactor design

Because of simplicity of operation, low operating costs and the ease with which liquid residence time can be varied, bubble-column slurry reactors have found a range of applications in gas-liquid-solid reaction systems [14]. This type of reactor holds the promise of improving the economics of methanol synthesis systems, owing to its ability to operate at very high conversions which are close to equilibrium levels, while maintaining essentially isothermal operation.

In an LPMeOH plant [4], synthesis gas containing CO, CO₂ and H₂ is passed upward into the reactor concurrent with the slurry which absorbs the heat liberated during reaction. The slurry is separated from the vapor and recirculated to the bottom of the reactor via a heat exchanger, where cooling occurs by steam generation. The reactor effluent gases are cooled to condense the products and any inert hydrocarbon liquid which may be vaporized. Methanol and the inert hydrocarbon liquid are immiscible and are separated by a decanter. The methanol stream produced is suitable for use directly or can be sent to a distillation unit to produce chemical-grade product. Unconverted gases are recycled back to the reactor. A small purge stream is taken off to limit the buildup of inert substances which may be present in the synthesis gas feed.

The primary advantages of this system over current technology are as follows.

(a) Due to the excellent reaction temperature control, high conversions of gas per pass can be economically realized such that the methanol concentration in the exit gas attains the 15 - 20 vol.% range as compared to a more normal figure of 2 - 6 vol.%. This in turn greatly reduces the recycle gas flow and compression requirement.

(b) The heat of reaction is largely recovered as high pressure steam in a simple manner.

(c) Reactor design is simplified in that liquids and gases are readily distributed across the reactor cross-sectional area without the

necessity for redistribution and quench along the reactor length.

(d) A small size catalyst can be used, thereby achieving higher rates of reaction with larger catalyst particles.

(e) Catalyst can be added and withdrawn from the system without the necessity of shutdown.

(f) The near isothermal temperature of the system permits optimum conditions favoring the desired reaction kinetics.

(g) Catalyst activity can be maintained at a constant "equilibrium" activity level so that it is not necessary to overdesign the reactor size for the "end of life" catalyst activity level.

In designing a slurry reactor, the goal is to optimize the height and diameter of reactor vessel for a specified capacity. First, the gas superficial velocity U_G is specified. The choice of U_G is influenced by several factors; too high a value could result in excessive gas holdup, whereas too low a value can result in uneconomically "hockey-puck-shaped" reactors. Given an allowable gas velocity and design synthesis gas feed flow, the necessary reactor cross-sectional area can be calculated [15].

With the diameter established, the reactor height (and hence the volume) depends on three basic factors: space velocity, slurry loading, and gas holdup. The choice of space velocity (here expressed as Nm³ of syngas treated per kg catalyst per hour) relates directly to reactor conversion through the kinetics and mass transfer correlations and this fixes the weight of the slurry loading, which is expressed in kg of catalyst per m³ of oil. A higher design slurry loading will always result in lower reactor volume other parameters being equal. However, too high a loading can increase the risk of plugging lines, valves and heat exchanger tubes.

Given the slurry volume and specified gas velocity and predicted gas holdup, the required reactor volume can be calculated.

2.2.1. Model assumptions

The model of methanol synthesis in the slurry phase is based on the following assumptions.

(i) The total pressure within the reactor is constant, *i.e.* the influence of hydrostatic head on gas expansion is neglected. Due to

the relatively high pressure used, this assumption should be reasonable.

(ii) There is no heat transfer to the surroundings, *i.e.* the reactor operates under adiabatic conditions.

(iii) The variation of gas flow rates across the column is neglected. The decrease in gas flow rate with an increase in conversion is assumed to counterbalance the increase in temperature along the reactor length.

(iv) For the flow pattern in the gas phase, a plug flow model is used. This corresponds to a general case in bubble column slurry reactors where the gas phase Bodenstein number is usually very high.

(v) The effectiveness factor for the pore diffusion inside the catalyst particles is taken as unity. Due to the very small particle size ($d_p < 100 \mu\text{m}$) used in a typical slurry reactor the diffusional limitations should be negligible.

(vi) The liquid-solid mass transfer resistances are neglected. In addition, the temperature difference between the catalyst and the liquid is assumed to be negligible.

(vii) The slurry phase is modelled by an axial dispersion model [16].

(viii) The catalyst distribution is not uniform over the entire suspension and it is modeled by the sedimentation-dispersion model [17, 18].

(ix) The hydrodynamic properties, *i.e.* gas holdup, interfacial area, heat and mass transfer coefficients and dispersion coefficients, are assumed to be spatially independent.

(x) Due to the low heat capacity of the gas phase compared to the slurry phase, only liquid and solid phases are considered in the energy balance.

(xi) The physico-chemical properties (*i.e.* density, viscosity, diffusivity, and solubility) are considered to be functions of temperature.

(xii) Both reactions, methanol formation and hydrogenation of CO_2 , are considered.

2.2.2. Model equations

Based on the above assumptions the following mass and energy balances can be written.

Mass balances for the i th component in the gas phase are represented as

$$\frac{d}{dx} (U_G C_{ig}) + (k_L a)_i (C_{ig}^* - C_{il}) = 0 \quad (12)$$

Scaling of this equation gives a dimensionless equation for gas phase mass balance

$$\frac{dy_i}{dz} + (1 + \theta)(\text{St}_G)_i (y_i - x_i) = 0 \quad (13)$$

In the liquid phase, for cocurrent operation, a differential mass balance yields

$$\begin{aligned} \frac{d}{dx} \left(\epsilon_L D_L \frac{dC_{iL}}{dx} \right) - \frac{d}{dx} (U_L C_{iL}) \\ + (k_L a)_i (C_{ig}^* - C_{il}) + \sum_k \nu_{ik} \epsilon_L C_{cat} r_k \\ = 0 \end{aligned} \quad (14)$$

where ν_{ik} is the stoichiometric coefficient for component i for reaction k . For products ν_{ik} is positive and for reactants it is negative. Again by scaling, we get the following dimensionless equation for liquid phase mass balance

$$\begin{aligned} \frac{1}{\text{Bo}_L} \frac{d^2 x_i}{dz^2} - q \frac{dx_i}{dz} + (\text{St}_L)_i (y_i - x_i) \\ + \sum_k \psi \text{Da}_k \phi_{ik} \\ = 0 \end{aligned} \quad (15)$$

Solid catalyst balance is represented on the basis of the sedimentation-dispersion model

$$D_s \frac{d^2 C_{cat}}{dx^2} + \left(U_{ss} - \frac{U_L}{1 - \epsilon_G} \right) \frac{dC_{cat}}{dx} = 0 \quad (16)$$

Integration with suitable boundary conditions results [19] in

$$\begin{aligned} \psi &= \frac{C_{cat}}{C_{cat, f}} \\ &= \frac{\text{Bo}_s \exp\{(\text{Bo}_s - \text{Bo}_L^*)(1 - z)\} - \text{Bo}_L^*}{\text{Bo}_s - \text{Bo}_L^*} \end{aligned} \quad (17)$$

Energy balance is dealt with as follows. Considering the slurry phase as pseudo-homogeneous, we can write the following balance for the adiabatic case

$$\begin{aligned} \frac{d}{dx} \left(\epsilon_L \lambda_{ax} \frac{dT}{dx} \right) - \frac{d}{dx} (U_G \bar{\rho} \bar{C}_p T) \\ + \sum_k \epsilon_L C_{cat} r_k (-\Delta H_R)_k \\ = 0 \end{aligned} \quad (18)$$

The dimensionless form of this equation is

$$\frac{1}{Pe} \frac{d^2\theta}{dz^2} - q \frac{d\theta}{dz} + \sum_k \psi Be_{ik} Da_k \phi_{ik} = 0 \quad (19)$$

Boundary conditions are set as follows. For $z = 0$

$$y_i = y_{i,f} \quad (20)$$

$$x_i = x_{i,f} + \frac{1}{Bo_L} \frac{dx_i}{dz} \quad (21)$$

$$\theta = \frac{1}{Pe} \frac{d\theta}{dz} \quad (22)$$

while for $z = 1$

$$\frac{dx_i}{dz} = \frac{d\theta}{dz} = 0 \quad (23)$$

2.2.3. Solution technique

The system of eleven differential equations (five for gas and five for liquid phase mass balances plus one for energy balance) represents the design model of the slurry-bed reactor and this system of equations is solved numerically using the method of orthogonal collocation. The computer program COLSYS [12] was again applied.

3. RESULTS AND DISCUSSION

3.1. Fixed-bed reactor

In the present study, the effects of operating and design conditions on the overall conversion and the space time yield (STY) are examined. All the calculations have been made for Lurgi gasifier-type syngas composition (see Table 1).

Some typical results illustrating the radial concentration profiles inside the pellet, and the concentration and temperature profiles

TABLE 1

Composition of syngas from a Lurgi gasifier

Component	Mole %
H ₂	50
CO	25
CO ₂	10
Inert substances	15

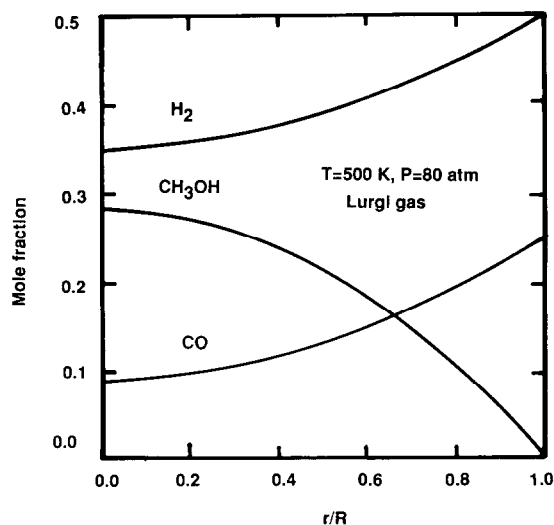


Fig. 1. A typical radial concentration profile plot for a catalyst pellet with a diameter of $d_p = 5$ mm. The temperature profile was found to be negligible under these conditions and is not included here. The calculations are carried out for Lurgi gas composition (Table 1).

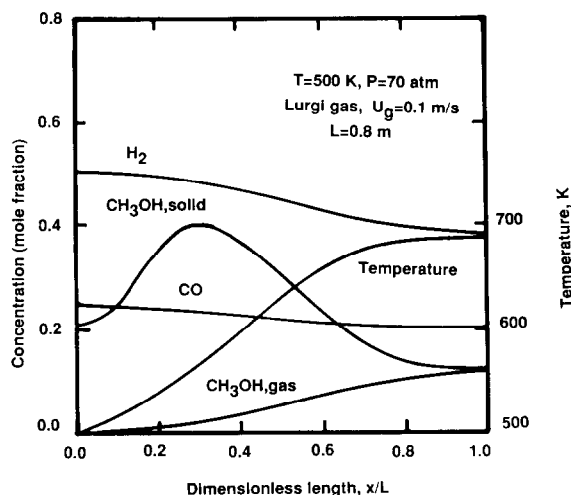


Fig. 2. Axial temperature and concentration profiles for a fixed bed reactor. Note that the catalyst center methanol concentration (CH_3OH , solid) goes through a maximum.

in the reactor are demonstrated in Figs. 1 and 2 respectively. In Fig. 1 only the mole fractions for CO, H₂ and CH₃OH are given and since temperature rise in the pellet was found to be very small (~ 1 K), it is not included there. It is clear from these figures that diffusional limitations inside the pellet are very strong under normal operating conditions and should play a very important role in the design calculations. External mass and energy transfer resistances, on the other

hand, were found to be negligible since the surface concentrations (mole fractions for H_2 , CO and CH_3OH are 0.5, 0.25 and 0 respectively) are equal to the bulk values. Under these conditions, the effectiveness factors (overall) are 0.87 and 0.43 for the methanol and shift reactions respectively.

In Fig. 2 several concentration and temperature profiles are shown for a fixed-bed reactor length of 0.8 m that corresponds to the first bed in ICI's quench reactor. For methanol, pellet center concentration is also included. As is shown in the figure the mole fraction of methanol increases in the bulk phase while it shows a maximum for the solid phase center concentration. The temperature increases vary rapidly and a temperature rise of 186 K is obtained. Actually this figure refers to a gas velocity of 0.1 m s^{-1} which is very low (a normal value is about 0.4 m s^{-1}) and such a temperature rise would not be permitted in a commercial reactor.

Overall effectiveness factors η_i are calculated from the effective reaction rates R_{eff} to signify internal and external mass transfer resistances

$$\eta_1 = \frac{R_{1,\text{eff}}}{R_{1,b}} = \frac{a_v D_M \left(\frac{dC_M^s}{dr} \right)_{r=R}}{(-r_{1,b} \rho_p)} \quad (24)$$

where $R_b = r_b \rho_p$ is the volumetric reaction rate calculated at bulk temperature and concentrations. In a similar fashion

$$\eta_2 = \frac{R_{2,\text{eff}}}{R_{2,b}} = \frac{a_v D_{CO_2} \left(\frac{dC_{CO_2}^s}{dr} \right)_{r=R}}{(r_{2,b} \rho_p)} \quad (25)$$

Typical overall effectiveness factors and effective reaction rates are given in Fig. 3. The overall effectiveness factor drops to 0.1 for the methanol synthesis reaction (η_1) while it behaves in a peculiar way for the shift reaction (η_2), namely, it goes through a slight maximum. The effective rates increase at first and then decrease with axial coordinate.

The overall conversion of syngas to methanol in a fixed bed reactor is defined in terms of the CO in the feed

$$X = \frac{N_M - N_M^o}{N_{CO}^o} = \frac{Q C_M - Q_o C_M^o}{Q_o C_{CO}^o}$$

where N_M and N_{CO} are the molal flow rates in the reactor for methanol and CO respec-

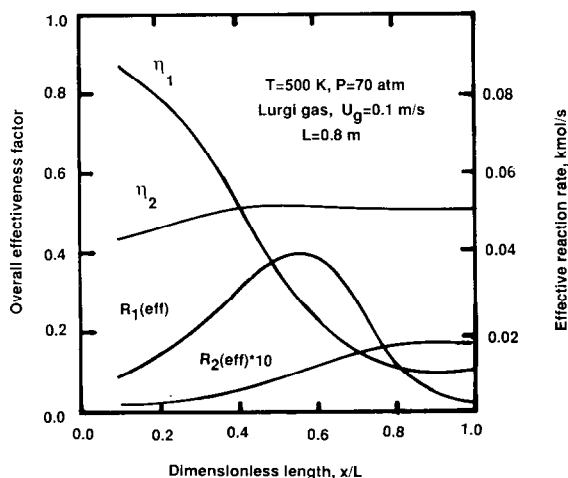
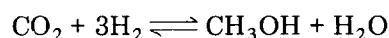


Fig. 3. Effective reaction rate and effectiveness factors for methanol synthesis and shift reactions. Here the effective reaction rate includes both external and internal diffusion resistances. The effective reaction rates are given for a catalyst particle of diameter 0.5 mm.

tively calculated as $N_i = Q C_i$, where Q is the volumetric gas velocity and C_i is the concentration. It should be noted that the conversion obtained in this manner has contributions from both methanol formation and reverse shift reactions (reactions 1 and 2 respectively). Methanol is produced not only from the CO that is provided in the feed stream but also from the CO obtained from the shift reaction. By adding reactions 1 and 2 we can represent the formation of methanol from CO_2



Therefore the actual conversion of CO in the feed X_1 needs to be calculated from the difference

$$X_1 = X - X_2$$

where X_2 is the conversion of shift reaction (reaction 2) given by

$$X_2 = \frac{N_{CO_2}^o - N_{CO_2}}{N_{CO}^o}$$

Increase in the gas velocity leads to a gradual drop in conversion as shown in Fig. 4. At this point it is desirable to define the STY as a performance index for the reactor. STY can be defined as the molal production rate of methanol (for instance in kmol h^{-1}) per kg of catalyst weight

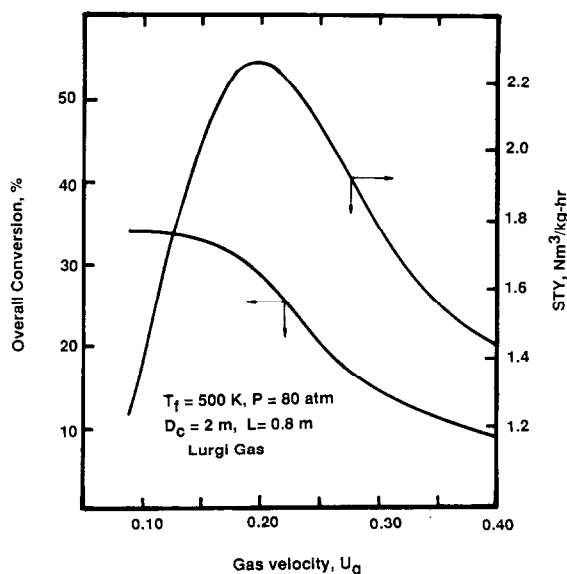


Fig. 4. Effect of gas velocity on conversion and STY in a fixed bed reactor.

$$(\text{STY})_M = \frac{N_M - N_M^0}{W}$$

where W is the weight of catalyst bed in the reactor. Space time yields defined in terms of volume at standard temperature and pressure are sometimes more appropriate and are used in this work

$$\text{STY} = V_M^{\text{stp}} (\text{STY})_M$$

where V_M^{stp} is the molal volume of syngas under standard conditions.

From Fig. 4 it may be seen that STY goes through a maximum. The higher the gas velocity, the lower the residence time and the lower the conversion. However STY directly increases with gas velocity (molal flow rate of methanol N_M in the equations above) and depends on the conversion. The optimum gas velocity under these conditions (500 K, 80 atm, $L = 0.8$ m) can be established as 0.18 m s^{-1} .

3.2. Slurry reactor

The model involves numerous quantities which may influence the performance of bubble column slurry reactor for methanol synthesis. In this part of the work a parametric study was performed to test the model and to investigate the reactor behavior under normal ranges of operating and design parameters. The computations were done for a large

scale reactor and the results are presented in graphical form.

The concentration profiles (mole fractions for gas phase and dimensionless concentration for liquid phase) are shown in Fig. 5 for a reactor 1 m in diameter and 8 m in length. The liquid phase profiles were found to be relatively flat, showing that dispersion is appreciable. The difference between gas and liquid phase concentration is small and it can be seen that the liquid phase is saturated after a dimensionless length of 0.2. As will be seen later, the methanol synthesis in the slurry reactor is kinetically controlled; hence the liquid phase concentration corresponds to the equilibrium (saturation) value.

Conversions and STY in a slurry reactor are defined in the same way as described for the fixed-bed reactor. However, since an additional (slurry) phase is involved, the molal flow rates N_i are calculated as follows

$$N_i = Q_g C_{ig} + Q_L C_{iL}$$

where Q_g and Q_L are the volumetric flow rates of gas and slurry in the reactor respectively.

The conversions for methanol formation and shift reaction are presented in Fig. 6. Both increase with axial coordinate and neither reaches equilibrium. As the rate of the shift reaction is very small, its contribution to the overall conversion is negligible.

The column diameter was varied between 1 and 4 m. The results are presented in Fig. 7

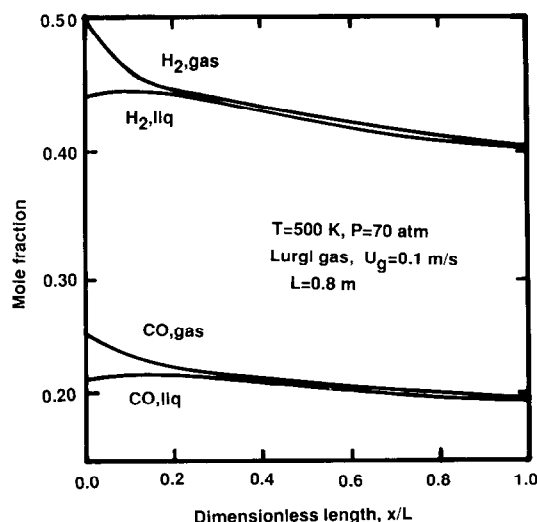


Fig. 5. Concentration profiles in a bubble column methanol reactor. Mole fractions and dimensionless liquid phase concentrations are used for gas and liquid phase concentrations respectively.

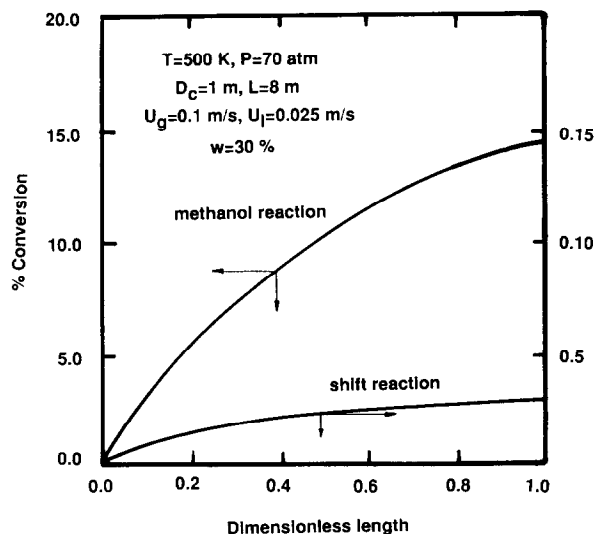


Fig. 6. Typical conversion profiles in a bubble column methanol reactor with Lurgi gas as a feed. Note that the conversion for shift reaction is negligible (0.03%).

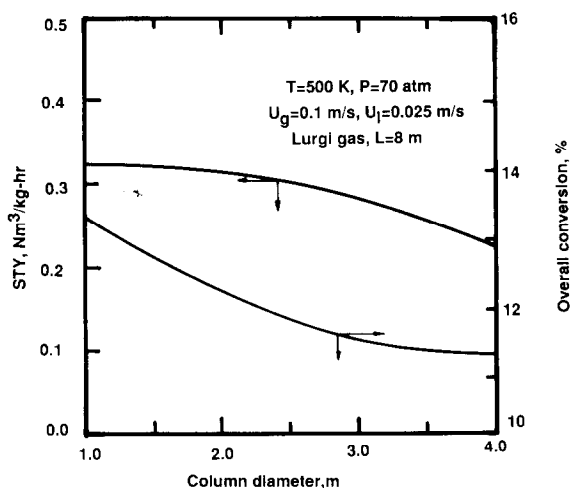


Fig. 7. Effect of column diameter on the performance of the bubble column methanol reactor.

for conversion and STY. A slight decrease is found when increasing the diameter, which is attributed to increased dispersion in the liquid phase. The overall effect is only moderate; however consideration of the effect may be very important in the scale-up of slurry reactors [20].

Increase in gas velocity leads to a gradual drop in conversion but also to a gradual increase in STY. The effects are described in Figs. 8 and 9 for different catalyst loadings. It is apparent that the reaction is kinetically controlled since conversion increases directly with catalyst concentration. At low gas velocities STY increases with decreasing

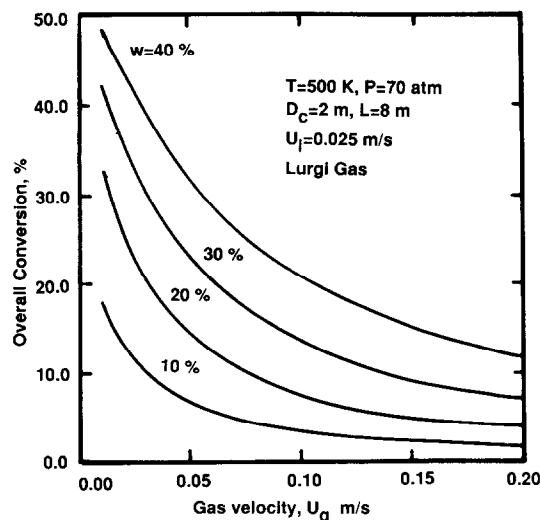


Fig. 8. Effect of gas velocity and catalyst loading on overall conversion in the bubble column reactor.

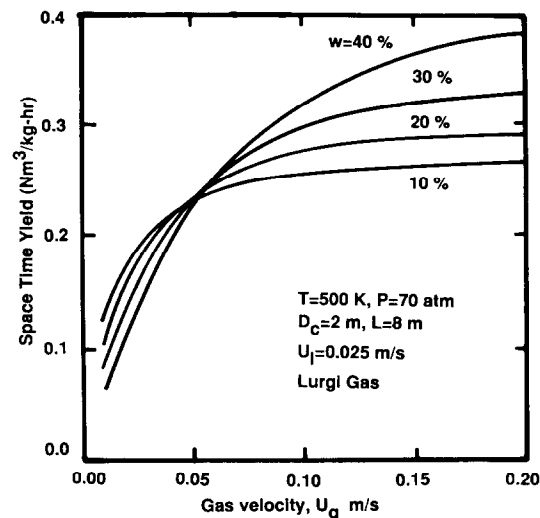


Fig. 9. Effect of gas velocity and catalyst loading on STY in the bubble column reactor.

catalyst loading, showing that mass transfer resistances can be important and the effect of catalyst loading increases the slurry viscosity loading with a decrease in mass transfer coefficients.

The slurry velocity was found to be a very important factor affecting performance. As may be seen from Fig. 10, both the conversion and STY increase drastically at low slurry flow rates, attaining a maximum at a critical slurry velocity. Here it must be pointed out that in a bubble column reactor all the design parameters are independent of the liquid velocity. The liquid velocity can only affect performance by altering the liquid phase

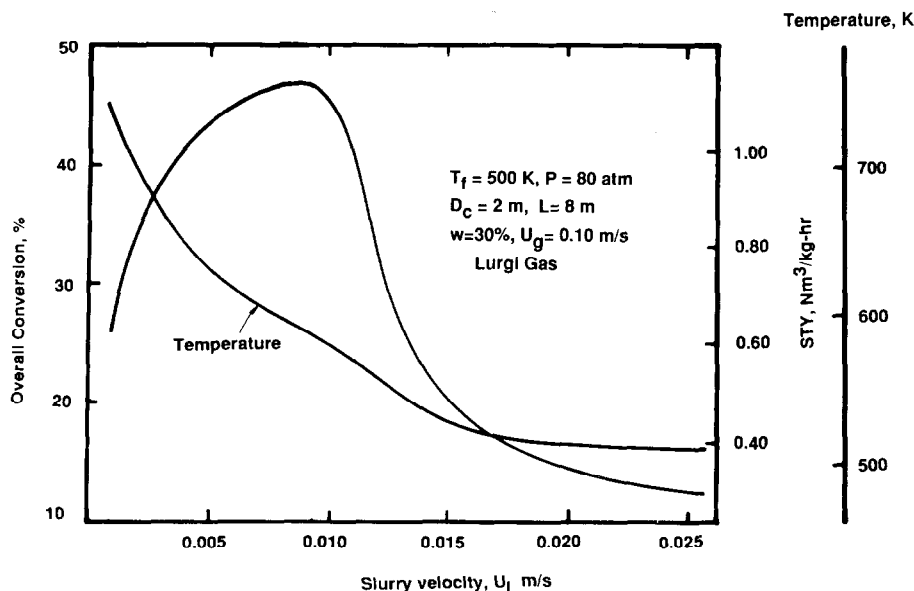


Fig. 10. Effect of slurry velocity on the performance of the bubble column methanol reactor. The temperature given in the figure is the outlet temperature.

(reaction phase) residence time and the rate of heat removal. The reactor temperature (here the outlet temperature is considered), also shown in Fig. 10, indicates that the latter is more important. As the liquid velocity decreases, the temperature of the reactor increases since the rate of heat removal decreases. The increase in the temperature directly depends on the reaction rate and since the synthesis is kinetically controlled it leads to a sharp increase in conversion and therefore STY. The increase in conversion goes through a maximum as the reaction is reversible and exothermic. The conversions obtained at higher temperatures (on the left of the maximum) correspond to equilibrium conversions. An optimal slurry velocity is estimated to be 0.9 cm s^{-1} under these conditions.

4. COMPARISON OF THE PERFORMANCES OF FIXED-BED AND SLURRY REACTORS FOR METHANOL SYNTHESIS

Based on reactor performance computations a comparison has been made of the two different reactor systems for methanol synthesis. Although a comprehensive comparison should also involve economic considerations, the present study deals only with reactor performance.

Fixed-bed reactors are preferred because of their ease of operation and the absence of a second phase (liquid), but they do have the drawback of temperature control. In fact, using a single bed it is not possible to achieve the desired conversion and the reactor should be designed in such a way that temperature control can be maintained. Operating conditions should also be carefully chosen to avoid equilibrium limitations. The reaction is found to be pore diffusion controlled; hence the reaction rate is lowered significantly.

Slurry reactors were introduced to provide a good solution for temperature control and to avoid pore diffusion by using a liquid and very fine catalyst particles. However, the presence of a second phase (liquid) introduced another problem in connection with mass transfer and solubility.

Generally speaking, each type of reactor has its own advantages and drawbacks. In the present study, the conversion and STY of the two types of reactor were compared. In the calculations the same feed conditions (composition, temperature, pressure) were used but in order to avoid equilibrium limitations, higher gas velocities and shorter bed lengths were used in the fixed-bed reactor. For the slurry reactor the catalyst loading was also varied.

Since our aim was to compare the performances of two types of reactor, we used the

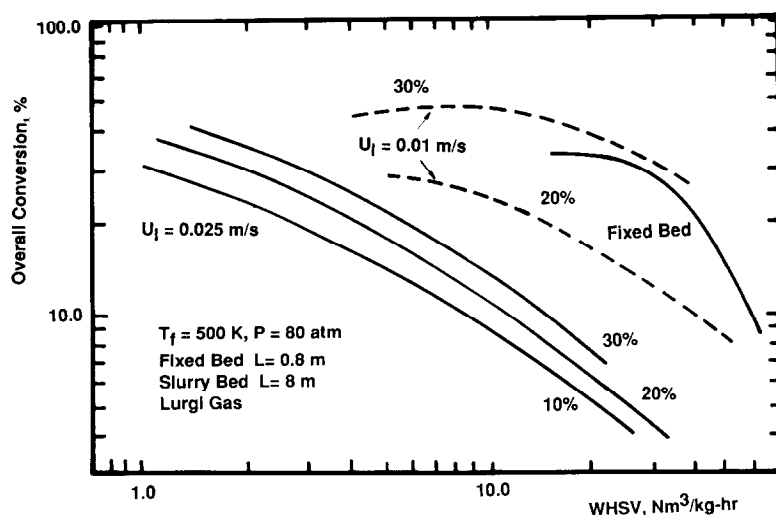


Fig. 11. Comparison of fixed and slurry bed methanol synthesis reactor performances on the basis of conversion.

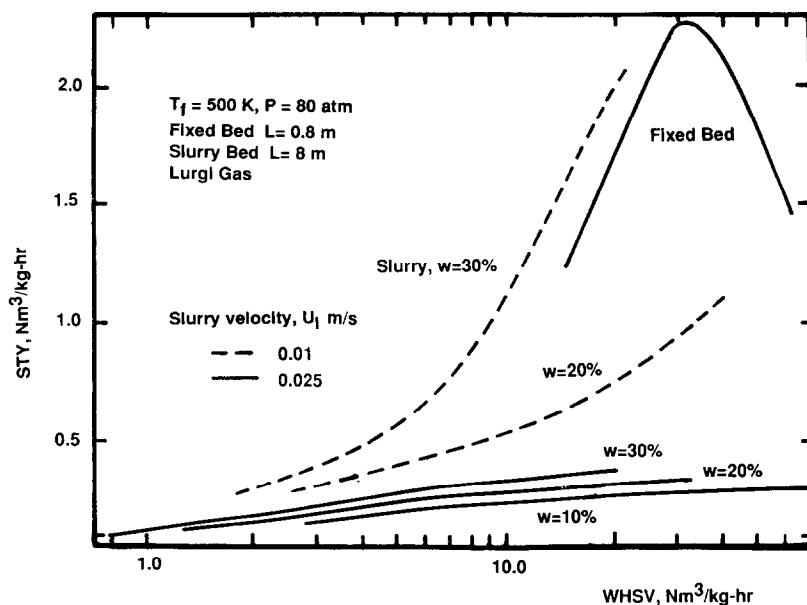


Fig. 12. Comparison of fixed and slurry bed methanol synthesis reactor performances on the basis of STY.

same reaction rate expression for both the reactors with the rate constants given for the *liquid phase*. Therefore the differences due to the alteration of rate constants have not been accounted for in the comparison of performances. As mentioned above, we have also used identical feed conditions so that only the differences in configuration and mode of operation between the reactors are explicitly reflected in their performance.

In Fig. 11 conversion is plotted *vs.* WHSV (weight hourly space velocity) for both reactors. It is clear that for the same WHSV slurry reactors can give higher conversions. This is mainly due to the alteration of equilib-

rium conversions by the presence of liquid phase. Figure 12 shows the STY for the two types of reactors. This figure shows that the slurry reactor can offer a better performance when operated under optimized conditions.

The results shown in Figs. 11 and 12 indicate that, in general, fixed- and slurry-bed reactors can give comparable performances when appropriate operating conditions are chosen. The performance of the fixed-bed reactors can be further improved by proper staging and interstage cooling of the reacting mixture. The slurry-bed reactor performance can also be further improved by choosing a better solvent for the reaction process.

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APPENDIX A: PARAMETER ESTIMATION

A.1. Fixed-bed reactor

A.1.1. Diffusivities

Binary diffusivities are calculated using the Fuller-Schetter-Giddins equation [A1]

$$D_{ij} = \frac{10^{-7} T^{3/2} (1/M_i + 1/M_j)^{1/2}}{P(v_i^{1/3} + v_j^{1/3})^2}$$

Gas mixture diffusivities are calculated by the method of Wilke [A2]

$$D_{i,m} = \frac{1 - y_i}{\sum_{j \neq i} y_j / D_{ij}}$$

For the effective diffusivities in catalyst pores, the parallel pore model of the catalyst structure is assumed and the contribution of Knudsen diffusivities is neglected

$$D_{i,m}^e = D_{i,m} \frac{\epsilon_p}{\tau}$$

A.1.2. Viscosities

The method of corresponding states [A1] is used to calculate the gas viscosities

$$\mu_i = \frac{(1.9T_{r,i} - 0.29)10^{-7} Z_{c,i}^{2/3}}{\zeta_i}$$

with

$$\zeta_i = \frac{T_{c,i}^{1/6}}{M_i^{1/2} P_{c,i}^{2/3}} \quad \text{and} \quad T_{r,i} = \frac{T}{T_{c,i}}$$

The viscosity of the methanol synthesis gas mixture is calculated by Wilke's equation [A1]

$$\mu_{\text{mix}} = \frac{\sum_{i=1}^n y_i M_i}{\sum_{j=1}^n y_j \phi_{ij}}$$

where

$$\phi_{ij} = \frac{1}{\sqrt{8}} \left(1 + \frac{M_i}{M_j} \right)^{-1/2} \left\{ 1 + \left(\frac{M_j}{M_i} \right)^{1/2} \left(\frac{\mu_i}{\mu_j} \right) \right\}^2$$

A. 1.3. Heat capacities

Heat capacities are taken from the literature in the form

$$C_{p,i} = a_i + b_i T + c_i T^2 + d_i T^3$$

The heat capacity of the methanol gas mixture is calculated by a simple mixing rule

$$C_{p,m} = \sum C_{p,i} y_i$$

The mass heat capacity of mixture is calculated by

$$C'_{p,m}(\text{mass}) = \sum \frac{C_{p,i} y_i}{M_i}$$

A.1.4. Conductivities

The thermal conductivity of each species is estimated by Bromley's method [A1]. The equations used are, for monoatomic gases

$$\frac{\lambda M}{\mu} = 2.5 C_v$$

for linear gases

$$\frac{\lambda M}{\mu} = 1.3 C_v + 3.50 - \frac{0.70}{T_r}$$

and for monoatomic gases

$$\frac{\lambda M}{\mu} = 1.30 C_v + 3.66 - 0.3 C_{ir} - \frac{0.69}{T_r} - 3\alpha$$

The thermal conductivity of the gas mixture is calculated using the Lindsay-Bromley equation [A1]

$$\lambda_m = \frac{\sum_{i=1}^n y_i \lambda_i}{\sum_{i=1}^n y_i A_{ij}}$$

where

$$A_{ij} = \frac{1}{4} \left[1 + \left\{ \frac{M_i}{M_j} \left(\frac{M_j}{M_i} \right)^{3/4} \frac{T + S_i}{T + S_j} \right\}^{1/2} \right]^2 \times \frac{T + S_{ij}}{T + S_i}$$

A.1.5. Mass transfer coefficients

The following correlation [9] is used for k_g values

$$J_D = \frac{0.458}{\epsilon_B} \left(\frac{d_p G}{\mu} \right)^{-0.407} = J_H = \frac{(k_g)_i}{U} \left(\frac{\mu}{\rho D_i} \right)^{2/3}$$

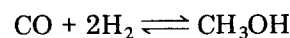
A.1.6. Heat transfer coefficient

Taking $J_D = J_H$, the h value is calculated from

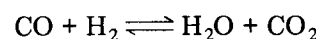
$$J_H = \frac{h}{C_p G} \left(\frac{C_p \mu}{\lambda} \right)^{2/3}$$

A.1.7. Reaction enthalpies

At 298 K



$$\Delta H_{298}^\circ = -94\,084 \text{ kJ kmol}^{-1}$$



$$\Delta H_{298}^\circ = 41\,270 \text{ kJ kmol}^{-1}$$

For any temperature using C_p values

$$\Delta H_T^\circ = \Delta H_{298}^\circ + \int_{298}^T \Delta C_p dt$$

A.1.8. Catalyst properties

The following properties are reported in the literature [A3] for the Cu/ZnO/Al₂O₃ catalyst: thermal conductivity, $\lambda_e = 4.18 \times 10^{-3} \text{ kJ m}^{-1} \text{ s}^{-1} \text{ K}^{-1}$; density, $\rho_p = 1980 \text{ kg m}^{-3}$; porosity, $\epsilon_p = 0.3$; tortuosity, $\tau = 7$.

A.1.9. Reaction kinetics

For the Cu/Zn/Cr₂O₃ catalyst (United catalyst T-2370) Berty *et al.* [A4] obtained the following rate expressions for the gas phase reaction

$$r_1 = k_1 \left(C_{\text{H}_2} - \frac{C_{\text{CH}_3\text{OH}}}{K_{\text{eq},1} C_{\text{H}_2} C_{\text{CO}}} \right)$$

and

$$r_2 = k_2 \left(C_{\text{H}_2} - \frac{C_{\text{CO}} C_{\text{H}_2\text{O}}}{K_{\text{eq},2} C_{\text{CO}_2}} \right)$$

The equilibrium constants $K_{\text{eq},1}$ and $K_{\text{eq},2}$ are calculated from the equations of Rosenman *et al.* [A5]. For the liquid phase reaction, a similar rate expression for r_1 is reported by Lee *et al.* [A6] with different activation energy and pre-exponential factor. Assuming that these parameters are affected by the presence of liquid for both reactions by the same factor, rate constants are set for r_2 in the slurry phase.

A.2. Slurry bed reactor

A.2.1. Physical properties of liquid phase

The following quantities are used for Witco-40 ($\text{C}_{18}\text{H}_{38}$): density [A1], $\rho_L = 620 \text{ kg m}^{-3}$ and assumed constant; viscosity [A1], $\mu = 8.8778 \times 10^{-6} 10^{777.4/T} \text{ kg m}^{-1} \text{ s}^{-1}$ with T in kelvins; surface tension [A6], $\sigma = 0.016 \text{ N m}^{-1}$.

A.2.2. Heat capacity

This was estimated for $\text{C}_{18}\text{H}_{38}$ using the group contribution method of Luria and Benson (ref. A1, p. 154): $C_p = 2.106 \text{ kJ kg}^{-1} \text{ K}^{-1}$.

A.2.3. Diffusivities of gases in liquid phase

Diffusivities are estimated using the Wilke-Chang equation (ref. A1, p. 567)

$$D_{i,B} = 1.173 \times 10^{-16} \frac{(\phi M_B)^{1/2} T}{\mu_B \nu_i^{0.6}} \text{ m}^2 \text{ s}^{-1}$$

A.2.4. Solubilities

Solubility and liquid-vapor equilibrium data for the species reported in the literature [A6] are expressed as Henry's constants

$$H_i = A_i \exp(B_i/T) \text{ atm m}^3 \text{ kmol}^{-1} \quad (\text{with } T \text{ in K})$$

A.2.5. Properties of suspension

Density is given by

$$\bar{\rho} = v_{\text{cat}} \rho_{\text{cat}} + (1 - v_{\text{cat}}) \rho_L \text{ kg m}^{-3}$$

where

$$v_{\text{cat}} = \frac{\rho_L W_{\text{cat}}}{\rho_{\text{cat}} - W_{\text{cat}}(\rho_{\text{cat}} - \rho_L)}$$

and

$$W_{\text{cat}} = \frac{M_{\text{cat}}}{M_{\text{sus}}}$$

i.e. the weight fraction of catalyst in suspension.

No correlation has been reported in the literature for the slurry viscosities of Witco-40 suspensions. Therefore, the correlation of Deckwer *et al.* [A7] is assumed to be applicable.

$$\bar{\mu} = \mu_L (1 + 4.5 V_{\text{cat}}) \text{ kg m}^{-1} \text{ s}^{-1}$$

Heat capacity is given by

$$\bar{C}_P = W_{\text{cat}} C_{p,\text{cat}} + (1 - W_{\text{cat}}) C_{p,L}$$

A.2.6. Mass transfer and hydrodynamic properties

Gas holdup [A8] is given by

$$\frac{\epsilon_G}{(1 - \epsilon_G)^4} = 0.2 \left(\frac{g D_c^2 \rho_L}{\sigma} \right)^{1/8} \left(\frac{g D_c^3}{\nu_L^2} \right)^{1/12} \times \frac{U_G}{(g D_c)^{1/2}}$$

Mass transfer coefficient [A9] is calculated using

$$(k_L a) \frac{D_c^2}{D_i} = 0.6 \epsilon_G^{1.1} \left(\frac{\nu_L}{D_i} \right)^{0.5} \left(\frac{g D_c^2 \rho_L}{\rho} \right)^{0.62} \times \left(\frac{g D_c}{\nu_L^2} \right)^{0.31}$$

A.2.7. Dispersion coefficients

In the liquid phase [20]

$$D_L = 0.768 U_G^{0.32} D_c^{1.34}$$

while the heat dispersion coefficient in the liquid is given by

$$\lambda_{\text{ax}} = D_L \bar{\rho} \bar{C}_p$$

A.2.8. Solid phase dispersion and sedimentation

The dispersion coefficient is given by

$$\text{Bo}_C = \frac{U_G D_c}{D_s} = \frac{13 \text{Fr}}{1 + 8 \text{Fr}^{0.85}}$$

with $\text{Fr} = U_G / (g D_c)^{1/2}$ (Kato *et al.*, 1972). The settling velocity in a particle swarm is calculated from

$$U_{\text{ss}} = 1.2 U_{\text{st}} \left(\frac{U_G}{U_{\text{st}}} \right)^{0.25} \left(\frac{1 - v_{\text{cat}}}{1 - v_{\text{cat}}^*} \right)^{2.5}$$

where $v_{\text{cat}}^* = v_{\text{cat}}$ at $C_{\text{cat}} = 0.1 \text{ g cm}^{-3}$. The settling velocity U_{st} of a single particle is calculated from

$$\text{Re} = \frac{\text{Ar}}{18} \quad \text{if } \text{Re} \leq 0.5$$

$$\text{Re} = \left(\frac{\text{Ar}}{13.9} \right)^{0.7} \quad \text{if } \text{Re} > 0.5$$

where

$$\text{Ar} = \frac{\rho_L(\rho_{\text{cat}} - \rho_L)g}{\mu_L^2} d_p^3$$

and

$$\text{Re} = \frac{U_{\text{st}} d_p}{\nu_L}$$

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APPENDIX B: NOMENCLATURE

a_v	gas-solid interfacial area in fixed bed (m^{-1})
Be	dimensionless energy group ($= (-\Delta H_R)P/(\bar{\rho}\bar{C}_p H_i T_f \nu_{ik})$)

Bo_L	liquid phase Bodenstein number ($= U_G L/(\epsilon_G D_L)$)
Bo_L^*	liquid phase Bodenstein number ($= U_G L/(D_s(1 - \epsilon_G))$)
Bo_s	solid phase Bodenstein number ($= U_{ss} L/D_s$)
$C_{i,j}, C^g$	gas phase concentrations (kmol m^{-3})
C_{ig}^*	equilibrium concentration in liquid phase ($= Py_i/H_i$) (kmol m^{-3})
C_{cat}	catalyst concentration ($= W\bar{\rho}$)
C_p	heat capacity ($\text{kJ kg}^{-1} \text{K}^{-1}$)
d_p	particle diameter (m)
D	diffusion coefficient ($\text{m}^2 \text{s}^{-1}$)
D_G	gas phase dispersion coefficient ($\text{m}^2 \text{s}^{-1}$)
D_L	liquid phase dispersion coefficient ($\text{m}^2 \text{s}^{-1}$)
D_s	solid phase dispersion coefficient ($\text{m}^2 \text{s}^{-1}$)
Da	Domköhler number ($= \epsilon_L C_{\text{cat},f} Lk/U_G$)
h	heat transfer coefficient ($\text{kJ m}^2 \text{s}^{-1} \text{K}^{-1}$)
H	Henry's constant ($\text{atm m}^3 \text{kmol}^{-1}$)
ΔH_R	reaction enthalpy (kJ kmol^{-1})
k	rate constant ($\text{m}^3 \text{kg}^{-1} \text{s}^{-1}$)
k_g	mass transfer coefficient (m s^{-1})
$k_L a$	volumetric mass transfer coefficient (s^{-1})
L	length (m)
M	molecular weight
N	molal flow rate (total) (kmol s^{-1})
P	pressure (atm)
Pe	Peclet number for heat transfer ($= U_G \bar{\rho} \bar{C}_p L/(\epsilon_{SL} \lambda_{ax})$)
q	flow rate ratio ($= U_1/U_G$)
r	radial coordinate (m)
r	reaction rate ($\text{kmol kg}^{-1} \text{s}^{-1}$)
R	particle radius (m)
R	universal gas constant
Re	Reynolds number ($= d_p G/\mu$)
St_G	Stanton number in gas phase ($= (k_L a) L R T_f / (U_G H_i)$)
St_L	Stanton number in liquid phase ($= (k_L a) L / U_G$)
T	temperature (K)
T^g	gas phase temperature (K)
T^s	solid phase temperature (K)
U, U_G	superficial gas flow rate (m s^{-1})
U_L	superficial liquid flow rate (m s^{-1})
U_{ss}	settling velocity of catalyst particles in swarm (m s^{-1})
x	axial coordinate (m)

x	dimensionless liquid phase concentration ($= HC_{il}/P$)
X	molal conversion
W	weight fraction of catalyst in suspension
y	mole fraction in gas phase
z	dimensionless axial coordinate ($= x/L$)

Greek symbols

ϵ	bed void fraction
ϵ_G	gas holdup
ϵ_L	liquid holdup
ϵ_p	particle porosity
η	overall effectiveness factor, eqns. (24) and (25)
θ	dimensionless temperature ($= (T - T_f)/T_f$)
λ	thermal conductivity ($\text{kJ m}^{-1} \text{s}^{-1} \text{K}^{-1}$)

λ_{ax}	heat dispersion coefficient ($\text{kJ m}^{-1} \text{s}^{-1} \text{K}^{-1}$)
ν	stoichiometric coefficient
ϕ	dimensionless rate ($= rH_i\nu/(k_k P)$)
ψ	dimensionless catalyst concentration ($= C_{cat}/C_{cat, f}$)
ρ	density (kg m^{-3})
ρ_p	particle density (kg m^{-3})

Subscripts and superscripts

e	effective
f	feed
g	gas
i	i th component
k	k th reaction
l	liquid
m	mixture
o	inlet
s	solid